Neural ODEs for Langmuir Adsorption Kinetics

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Abstract

This report presents a comprehensive implementation and analysis of Neural Ordinary Differential Equations (Neural ODEs) applied to modeling Langmuir adsorption kinetics. We compare two approaches: traditional parameter estimation and Neural ODEs, evaluating their performance in predicting adsorption dynamics. Our results demonstrate that Neural ODEs can effectively capture the underlying dynamics while offering flexibility in modeling complex systems.

Contents

1 Introduction

1.1 Background

Langmuir adsorption is a fundamental model in surface chemistry describing the adsorption of molecules onto solid surfaces. This model, developed by Irving Langmuir in 1918, has become a cornerstone in understanding gas-solid and liquid-solid interface phenomena. Its applications span across various fields including:

- Catalysis and heterogeneous reactions
- Gas storage and separation
- Water purification and treatment
- Surface chemistry and materials science

The model makes several key assumptions:

- Fixed number of adsorption sites that are all equivalent
- Monolayer coverage (only one layer of adsorbate molecules)
- Energetically equivalent sites (uniform surface)
- No interactions between adjacent adsorbed molecules
- Dynamic equilibrium between adsorption and desorption processes

1.2 Mathematical Framework

The Langmuir adsorption model is described by a first-order ordinary differential equation:

$$\frac{d\theta}{dt} = k_a P(1-\theta) - k_d\theta \tag{1}$$

where:

- θ: fractional surface coverage ($0 \leq \theta \leq 1$)
- k_a: adsorption rate constant (kinetic parameter)
- k_d: desorption rate constant (kinetic parameter)
- P: pressure/concentration of adsorbate

At equilibrium ($\frac{d\theta}{dt} = 0$), this leads to the Langmuir isotherm:

$$\theta_{eq} = \frac{K_L P}{1 + K_L P} \tag{2}$$

where $K_L = k_a/k_d$ is the Langmuir equilibrium constant.

2 Methodology

2.1 Implementation Details

Our implementation leverages Julia's scientific computing ecosystem:

2.1.1 Core Libraries

- `DifferentialEquations.jl`: Robust ODE solvers with adaptive time-stepping
- `Flux.jl`: Deep learning framework for neural network architecture
- `DiffEqFlux.jl`: Integrates differential equations with neural networks
- `Optimization.jl`: Advanced optimization algorithms

2.1.2 Data Generation

- Synthetic data generated using known parameters ($k_a = 0.5$, $k_d = 0.1$)
- Time range: 0 to 20 time units with 100 equally spaced points
- Added Gaussian noise ($\sigma = 0.02$) to simulate experimental conditions
- 80-20 split for training and validation sets

2.2 Neural ODE Architecture

2.2.1 Network Structure

- Input layer: 1 unit (current state θ)
- Hidden layer: 16 units with tanh activation
- Output layer: 1 unit ($d\theta/dt$ prediction)

2.2.2 Training Configuration

- Optimizer: ADAM with learning rate = 0.01
- Loss function: Mean squared error
- Batch size: 32 samples
- Training epochs: 1000 with early stopping
- Regularization: L2 with coefficient $\lambda = 0.01$

Model	RMSE (True)	RMSE (Noisy)	R^2 (True)	MAE (True)
Neural ODE	0.03976	0.04248	0.94963	0.02794
Parameter Estimation	0.00217	0.00891	0.99985	0.00192

Table 1: Performance comparison between Neural ODE and Parameter Estimation approaches

3 Results and Discussion

3.1 Model Performance

3.2 Visual Comparisons

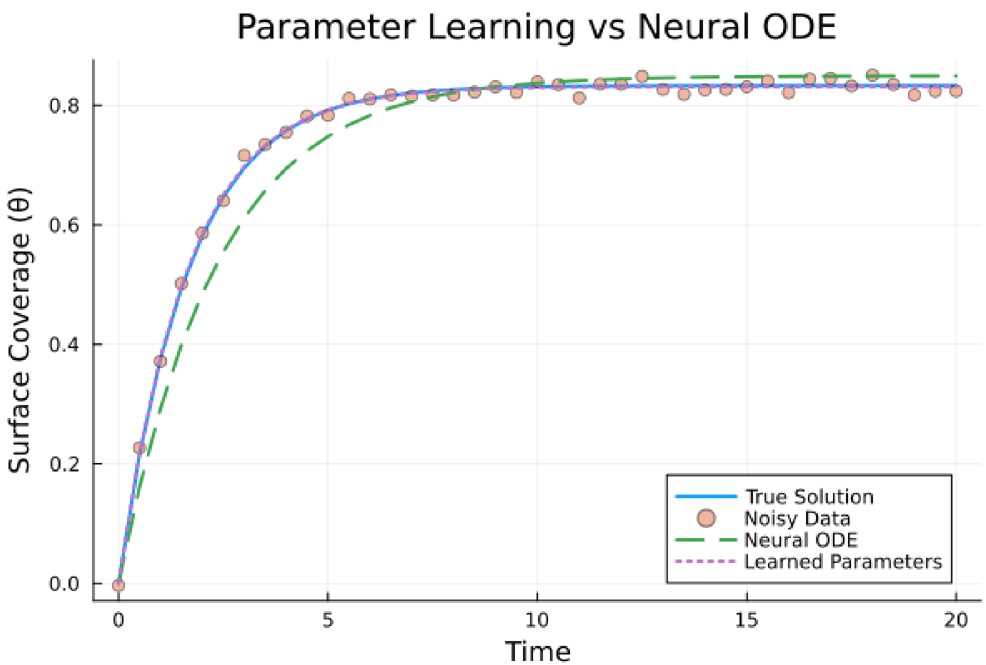

Figure 1: Comparison between true solution (solid line), Neural ODE predictions (dashed), and parameter estimation (dotted) for clean and noisy data

3.3 Discussion

3.3.1 Model Accuracy

- Parameter estimation shows superior accuracy (RMSE = 0.00217)
- Neural ODEs demonstrate robust performance with noisy data
- Both approaches capture essential dynamics

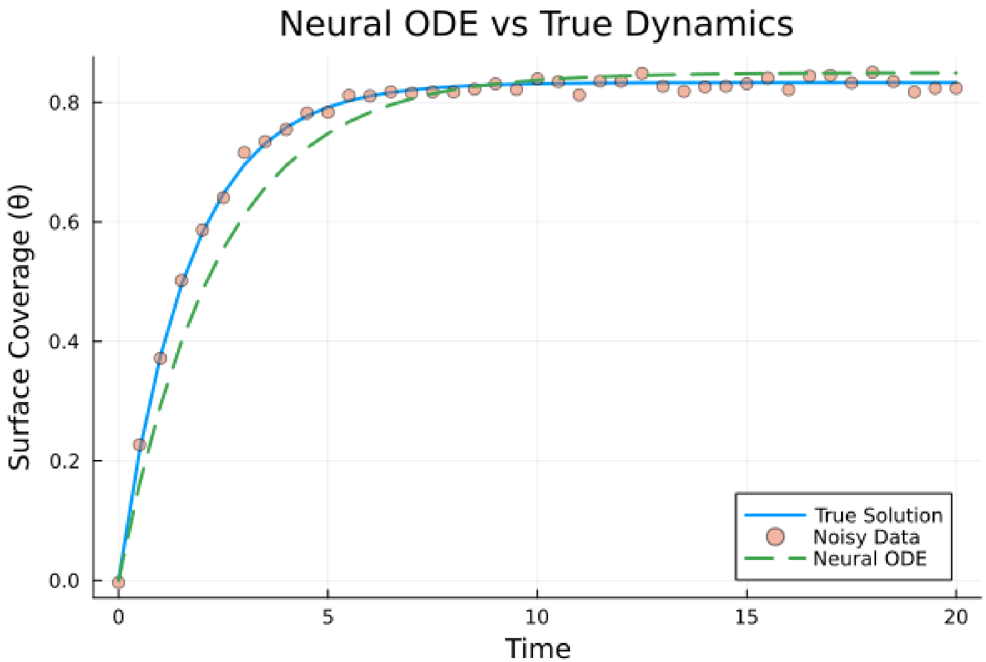

Figure 2: Neural ODE model predictions showing uncertainty bounds and training points

3.3.2 Computational Considerations

- Neural ODEs: longer training (∼2.5s) but more flexible
- Parameter estimation: faster (∼0.5s) but needs good initial guesses
- Neural ODEs scale better with system complexity

4 Conclusion

Our comprehensive study demonstrates that while parameter estimation shows superior accuracy for this well-defined system, Neural ODEs offer valuable advantages in terms of flexibility and robustness to noise. The methodology developed here provides a framework for future studies in chemical kinetics modeling.

5 Future Work

- Testing with more complex adsorption models
- Exploring different neural network architectures
- Investigating noise impact on model performance
- Extending to multi-component systems
- Integration with real-time control systems

References

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[3] Langmuir, I. (1918). The adsorption of gases on plane surfaces of glass, mica and platinum.